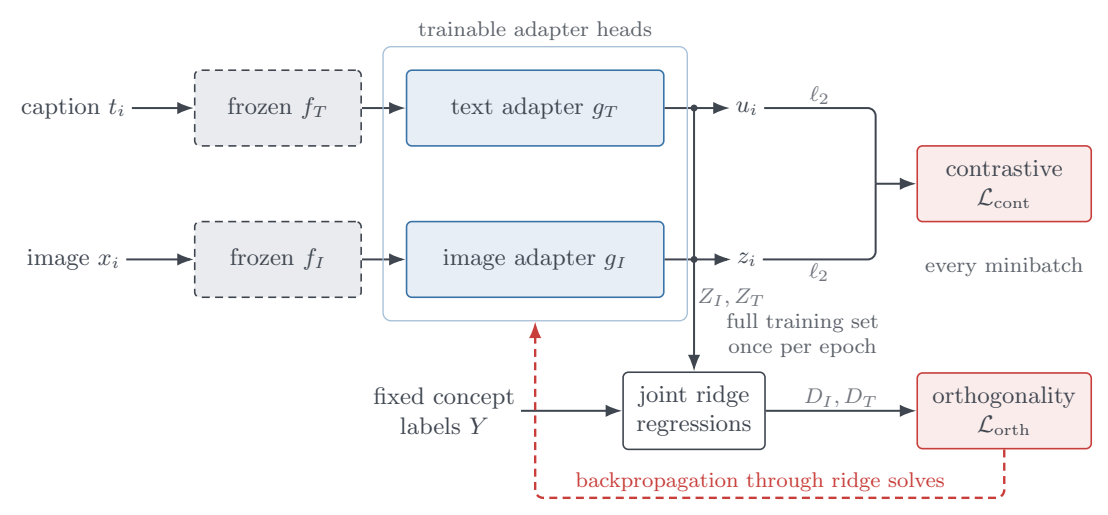

trainable adapter heads
caption t_i
frozen f_T
text adapter g_T
u_i
ℓ_2
contrastive $\mathcal{L}_{\text{cont}}$
image x_i
frozen f_I
image adapter g_I
z_i
ℓ_2
every minibatch
Z_I, Z_T
full training set
once per epoch
fixed concept labels Y
joint ridge regressions
D_I, D_T
orthogonality $\mathcal{L}_{\text{orth}}$
backpropagation through ridge solves